MARCEL: Multifaceted SpAtial-TempoRal ContrastivE Learning for Generic Spatial-Temporal Representations

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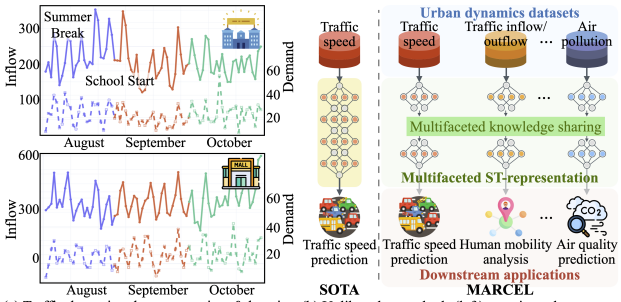
***Abstract*—The development of sensing technologies has broadened the scope of urban dynamics research. However, existing methods primarily focus on using isolated aspects of urban data, limiting their ability to capture the complex spatial-temporal dependency among different urban dynamics and generalizability across applications. Addressing these shortcomings requires a more comprehensive model capable of integrating multifaceted data, generating generalized representations adaptable to diverse applications and scenarios. In this paper, we introduce the Multifaceted SpAtial-TempoRal ContrastivE Learning framework, *i.e.*, MARCEL, an innovative approach designed to learn robust, universally applicable, and adaptable representations of multifaceted urban dynamics through contrastive learning. MARCEL employs pretrained preliminary representation learning modules to extract distinct spatial-temporal dependencies inherent to each urban dynamic independently. It then features a Spatial-Temporal Contrastive Learning strategy to capture unified spatial-temporal patterns, including asynchronous, conflicting, and complementary behaviors across multifaceted urban dynamics. Additionally, MARCEL integrates a Multifaceted Knowledge Transfer mechanism to capture inter-dependencies among different urban dynamics and facilitate knowledge sharing. The learned representations are highly generalizable and can be applied effectively to various downstream tasks. Extensive experiments on real-world urban datasets demonstrate that MARCEL is effective and significantly outperforms state-of-the-art baselines.**

***Index Terms*—Contrastive Learning, Representation Learning, Urban Dynamics Prediction, Spatial-Temporal Data Mining**

I. Introduction

Urban dynamics describes the interaction between spatial structures and temporal patterns in cities, providing insights into mobility trends and urban planning. It encompasses diverse phenomena, including traffic flows, human mobility, travel demand, air quality [63], *etc*. A comprehensive analysis of urban dynamics is crucial for various applications, such as traffic prediction, transit system design, resource distribution, public safety, and location-based services (*e.g.*, route optimization and ride-sharing) [34], [68].

Limitations of Existing Works. Numerous studies have sought to solve diverse urban problems using urban dynamics data. Many of these works [55], [69], [41], [8], [54], [30], [19], [53], [67], [63], [57] apply deep neural networks to predict urban dynamics, particularly for traffic prediction to improve traffic conditions. Additionally, generative model-based innovations [59], [60], [62], [61] focus on traffic generation problems under specific conditions, helping urban planning. Furthermore, other studies have developed models tailored for specific tasks such as human mobility prediction [28], [51], air quality forecasting [66], [20], *etc*. While effective for specific problems, these approaches are typically context-based, relying on distinct models and datasets tailored to individual scenarios, which limits their generalizability across multiple applications.



(a) Traffic dynamics change over time & location with human activity. (b) Unlike other methods (left), we aim to learn representations from multifaceted urban data (right).

Fig. 1: Spatial-temporal heterogeneity (a) and our proposed MARCEL framework comparison with SOTA approaches (b).

Representation learning has demonstrated its potential in learning generalizable representations that perform well across diverse downstream applications, particularly for sequential data [39], [36], [31], [50], [48] and images [9], [21], [5], [42], [35], [37], [40]. However, representation learning for spatial-temporal urban dynamics yields limited attention. Existing approaches primarily focus on one specific urban dynamic type or a class of applications. For instance, pre-trained models have been developed to learn representations for human-generated urban trajectories [16], [58], [17], while Wang et al. [47] proposed a framework for urban POI recommendations. MVGCL [65] learns urban region representations from human trajectories, and ST-GAN [56] introduces a disentangled representation learning framework for interpreting and predicting traffic flow. However, these methods are tailored

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to a single type of urban data, overlooking the dependencies of multifaceted urban dynamics. None of them is capable of generating generic representations for multifaceted urban dynamics that are applicable to diverse urban applications.

Goal and Challenges. This paper aims to develop a universal spatial-temporal representation model that can flexibly and comprehensively learn from various facets of urban dynamics, thereby producing versatile representations. These representations should be adaptable to a wide range of downstream applications, including traffic speed and congestion prediction, and human mobility analysis, *etc*., as shown in Fig. 1(b). Achieving this goal requires overcoming two critical challenges:

C1: *Different urban dynamics exhibit unique spatial-temporal patterns, which may be asynchronous, conflicting, or complementary.* For example, Fig. 1(a) illustrates how traffic inflow patterns evolve over three months in a school area, while these patterns differ significantly around a shopping mall. Furthermore, travel demand exhibits distinct spatial-temporal patterns compared to traffic inflow. These examples highlight that each type of urban dynamic is characterized by unique spatial-temporal patterns. The complexity increases further as these various urban dynamics being asynchronous, conflicting, or complementary. As a result, capturing these diverse and intricate spatial-temporal patterns across different urban dynamics and integrating them into a unified representation learning framework remains a significant challenge.

C2: *Urban dynamics are multifaceted and interdependent.* A city's transportation system consists of a complex network of transit modes, influenced by factors such as human mobility, air quality, and power consumption. These dynamics are highly interdependent, where changes in one aspect can propagate across others. A flexible and comprehensive model is needed to learn multifaceted urban dynamics while effectively capturing their interconnections.

Our Approach. To address the challenges outlined above, we propose the Multifaceted SpAtial-TempoRal ContrastivE Learning (MARCEL) framework. This innovative framework is designed to learn robust, universally applicable, and adaptable representations from multifaceted urban dynamics data using a novel contrastive learning approach. *Our contributions are summarized as follows:*

- To the best of our knowledge, MARCEL is the first contrastive learning framework specifically designed to learn generic and robust representations across diverse types of urban dynamics data. These representations can be effectively leveraged in a wide range of downstream applications, such as traffic speed prediction and urban dynamics generation (Sec. III).
- In MARCEL, we introduce a pretrained preliminary representation learning module to capture the distinct spatial-temporal dependencies inherent to each urban dynamic. This is followed by a novel Spatial-Temporal Contrastive Learning strategy, which incorporates spatial and temporal contrastive losses to identify and capture unified spatial-temporal patterns across multifaceted urban dynamics. Furthermore, MARCEL integrates an innovative Multifaceted Knowledge Transfer mechanism by introducing a pairing contrastive loss. This mechanism effectively captures the interdependencies among multifaceted urban dynamics across diverse geographical regions and over time. These advancements enable MARCEL to adaptively account for geographical and temporal variations, producing tailored representations suitable for a wide range of downstream tasks (Sec. III).
- We conduct extensive experiments on five real-world urban dynamics datasets across three cities to evaluate the effectiveness of MARCEL. The experimental results demonstrate that MARCEL outperforms other state-of-the-art representation learning methods, and the representations it extracts significantly enhance various downstream applications. *We have made our code and unique datasets available to the research community via an GitHub link* [1] (Sec. IV).

II. Overview

A. Preliminaries

Contrastive Learning. There has been a rich literature trying to capture the complex data patterns and learn useful representations from sequential data via contrastive learning [39], [36], [31], [50], [48]. For a given data point $x \in X$, the objective is to generate positive samples x^+ through data augmentation and negative samples x^- that differ in context or meaning. The standard contrastive learning aims to train a representation model f such that it brings the representations of positive samples, $(f(x), f(x^+))$, closer together, while pushing the representations of negative samples, $(f(x), f(x^-))$, further apart. *i.e*.,

$$\|f(x)-f(x^+)\|_2^2+\alpha < \|f(x)-f(x^-)\|_2^2, \ \ \forall(x,x^+,x^-) \in \mathcal{T}.$$

Here, α is a margin that is enforced between positive and negative pairs, and $\mathcal{T}$ is the set of all possible triplets in the training data. When generating the positive samples, a lot of works mainly rely on approaches such as random cropping, rotation, color jittering, or noise addition for image data [9], [21], [5], [42], [35], or semantic alignment, event alignment, time-wrapping for sequential data [44], [18], [3], [49], [1]. However, none of these approaches consider the complicated spatial-temporal correlations and multifaceted data sources at the same time.

B. Problem Definition

Definition 1 (A grid cell s_{ij}). To attain a more granular understanding of urban dynamics, we partition an urban area (*e.g*., a city) into $m \times n$ grid cells, each grid cell has uniform dimensions (*e.g*., $1\times1km^2$). The set of grid cells is represented as $\mathcal{S} = \{s_{ij}\}$, where $1 \le i \le m$, $1 \le j \le n$.

Definition 2 (A target region R_{ij}). For each target grid cell s_{ij}, we define a corresponding target region R_{ij}. R_{ij} is a square geographic area, encompassing $\ell\times\ell$ grid cells, with the cell s_{ij} positioned at its center. Formally, we represent a target region as $R_{ij} = \langle s_{ij}, \ell\rangle$. In our analysis, we operate under the

[1] MARCEL code: https://github.com/Yliu1111/Marcel.git.

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premise that there exists a significant spatial correlation among the urban dynamics of grid cells within the same target region R_{ij}.

Definition 3 (Urban dynamic feature $\boldsymbol{X}^k$). Urban dynamics encompass a wide range of features such as traffic speed, vehicle inflow and outflow, human mobility, *etc*. We consider the urban dynamic feature of type k for a specific grid cell s during a given time interval t to be represented by a scalar value x_t^k. The collective features of type k from all grid cells within a designated target region R at time t constitute a feature map for that region. This feature map is represented as a matrix $\boldsymbol{X}_t^k \in \mathbb{R}^{\ell \times \ell}$, and there are K distinct types of features, thereby we have $1 \leq k \leq K$.

Definition 4 (Urban dynamic history $\mathcal{D}^k$). Urban dynamics observations $\mathcal{D}^k$ include the data for an urban dynamic feature k over a span of time for the target region R, specifically, $\mathcal{D}^k = \{\boldsymbol{X}_t^k\}_{t=1}^T$, where T denotes the series length of $\mathcal{D}^k$. For any given time slot t, $\mathcal{D}_t^k$ represents the accumulated historical observations up to (and including) that time slot, formalized as $\mathcal{D}_t^k = \{\boldsymbol{X}_1^k, \cdots, \boldsymbol{X}_t^k\}$.

Problem Definition (Multifaceted Spatial-Temporal Representation Learning for Urban Dynamics). Given K different types of historical urban dynamics data $\{\mathcal{D}^1, \cdots, \mathcal{D}^K\}$, we aim to train a representation model f parameterized by θ. The representation model $f_\theta : \mathbb{R}^{\ell \times \ell} \to \mathcal{Z}$ can transform multifaceted urban dynamics features $\{\boldsymbol{X}^1, \boldsymbol{X}^2, \cdots, \boldsymbol{X}^k\}$ into a latent representation space $\mathcal{Z}$, *i.e*., $\boldsymbol{z}^k = f_\theta(\boldsymbol{X}^k)$ and $\boldsymbol{z}^k \in \mathcal{Z}$ when $k = 1, \cdots, K$. The learned representations for each urban feature type are universal and generic to be utilized in diverse downstream applications.

III. Methodology

In this section, we solve the multifaceted spatial-temporal representation learning problem for urban dynamics by introducing MARCEL — a novel Multifaceted Spatial-Temporal Contrastive Learning framework. MARCEL integrates a pretrained preliminary representation learning module alongside an innovative spatial-temporal contrastive learning strategy to capture both the unique and unified spatial-temporal patterns of various urban dynamics (*i.e*., tackling **C1**, see Sec. III-A and Sec. III-B). Additionally, MARCEL includes a multifaceted knowledge transfer mechanism, featuring a knowledge pairing contrastive loss incorporated into the final objective. This module facilitates effective knowledge sharing across different urban dynamics datasets (*i.e*., tackling **C2**, see Sec. III-C).

A. Pretrained Preliminary Representations

Multifaceted urban dynamics data are inherently complex, as each type of data (*e.g*., traffic flow, travel demand, air quality) exhibits distinct spatial-temporal patterns and dependencies. Directly training a single representation learning model to handle all these diverse dynamics simultaneously often leads to diluted representations that fail to capture the specific characteristics of each dynamic. Thus, we first design a Pretrained Preliminary Representation Learning Module that independently learns preliminary representations for each type of urban dynamics data. By focusing on a simple spatial-temporal prediction task for each type of data, this module ensures that the representations are rich, type-specific, and optimized for capturing the unique dependencies of each dynamic. These preliminary representations form a robust foundation for subsequent integration into the contrastive learning framework.

This lightweight module is implemented using a few layers of ConvLSTM [41]. For each type of urban dynamics data, denoted as $\mathcal{D}^k = \{\boldsymbol{X}_t^k\}_{t=1}^T$, the module is trained to perform a simple spatial-temporal prediction task. Specifically, the model predicts the next time step $\boldsymbol{X}_{t+1}^k$ based on previous observations. The loss function for each type of urban dynamics is defined as:

$$\mathcal{L}_{\text{pretrain}}^k = \frac{1}{T-1} \sum_{t=1}^{T-1} \left\| \hat{\boldsymbol{X}}_{t+1}^k - \boldsymbol{X}_{t+1}^k \right\|_2^2, \tag{1}$$

where $\hat{\boldsymbol{X}}_{t+1}^k$ is the predicted output of the ConvLSTM model for urban dynamic k, and $\boldsymbol{X}_{t+1}^k$ is the true spatial-temporal data at time step $t+1$. Pretraining separate modules for each type of urban dynamic data allows the model to specialize in capturing the unique spatial-temporal dependencies of its respective urban dynamic without interference from other dynamics.

Once the separate modules are trained, the final hidden layer outputs are extracted as the preliminary representations of the original urban dynamics data for each type. This transforms the initial data $\mathcal{D}^k = \{\boldsymbol{X}_t^k\}_{t=1}^T$ into preliminary latent representations $\tilde{\mathcal{D}}^k = \{\tilde{\boldsymbol{X}}_t^k\}_{t=1}^T$. These latent representations preserve the unique spatial-temporal dependencies inherent to each type of urban dynamic while reducing the complexity of the original data. The subsequent contrastive learning framework then operates on these preliminary latent representations.

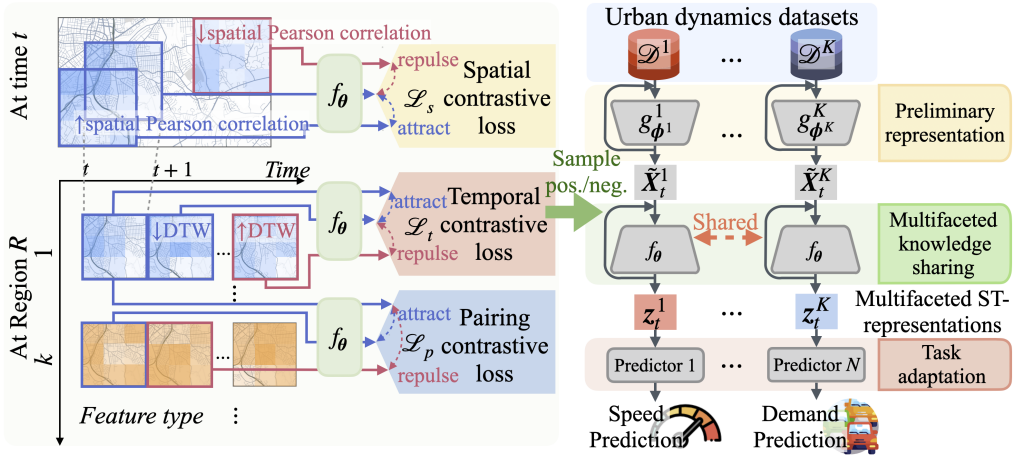
B. Spatial-Temporal Contrastive Learning

After learning independent preliminary representations for each type of urban dynamic, we effectively preserve their distinct spatial-temporal patterns. However, these patterns can be asynchronous, conflicting, or complementary when considered together. Without explicitly addressing these complexities, inconsistencies may arise, as the separate pretrained models fail to align or reconcile such disparities. This highlights the need for a more advanced approach to accurately capture and unify these patterns within a cohesive representation framework.

Spatial-temporal contrastive loss. We propose a spatial-temporal contrastive learning strategy to effectively capture the unified spatial-temporal patterns (*e.g*., including conflicting and complementary patterns) for multifaceted urban dynamics by generating positive and negative samples and designing contrastive losses from two key perspectives: spatial variation and temporal dynamics.

• *Spatial contrastive loss* is designed to capture the unified impact of geographical variations on multifaceted urban dynamics by leveraging both geographical proximity and functional

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(a) Spatial, temporal and pairing contrastive losses. (b) MARCEL solution framework.

Fig. 2: Structure of the MARCEL framework.

similarity. Viewing the multifaceted urban dynamics within a specific region as a cohesive whole allows us to better understand and model the combined spatial effects of various dynamics in that region. Regions closer in geographical distance are typically expected to share similar geographical features. Therefore, we select candidate positive samples as multifaceted urban dynamic features from regions near the target region, emphasizing their spatial similarities, while candidate negative samples are chosen as features from regions farther away, highlighting the contrasts in their spatial patterns. However, geographical proximity alone is not sufficient, as neighboring regions may serve different purposes and exhibit distinct characteristics. For example, a business district and a residential area might be geographically close but functionally distinct. Therefore, we refine the candidate positive and negative samples using spatial Pearson correlation [6]. As shown in Figure 2(a), among the candidate positive samples, regions with high spatial Pearson correlation with target region are defined as positive samples, while candidate negative samples with low spatial Pearson correlation are confirmed as negative samples. Therefore, given concatenated K types multifaceted urban dynamic features $\tilde{\boldsymbol{X}} = \text{concat}(\tilde{\boldsymbol{X}}^1, \tilde{\boldsymbol{X}}^2, \cdots, \tilde{\boldsymbol{X}}^K)$ at time t and target region R, positive samples $\tilde{\boldsymbol{X}}^{s+}$ are features recorded at the same time t, from nearby regions whose features have high spatial Pearson correlation with R. Negative samples $\tilde{\boldsymbol{X}}^{s-}$ are features recorded at the same time t, from distant regions with low spatial Pearson correlation with R. Therefore, the spatial contrastive loss tries to maximize the margin between positive spatial pairs and negative ones, *i.e.*,

$$\mathcal{L}_s \triangleq \sum_{\substack{\tilde{\boldsymbol{X}}, \tilde{\boldsymbol{X}}^{s+}, \\ \tilde{\boldsymbol{X}}^{s-} \in \tilde{\mathcal{D}}}} \max(0, \|f(\tilde{\boldsymbol{X}}) - f(\tilde{\boldsymbol{X}}^{s+})\|_2^2 - \|f(\tilde{\boldsymbol{X}}) - f(\tilde{\boldsymbol{X}}^{s-})\|_2^2 + \alpha_s), \tag{2}$$

where α_s represents the margin between positive and negative spatial contrastive pairs.

Algorithm 1 MARCEL Training Procedure

Input: Dynamics data $\{\mathcal{D}^k\}$; initialized $\{g_{\phi^k}^k\}$, f_θ.
Output: Well-trained $\{g_{\phi^k}^k\}$ and f_θ.

```
for urban dynamics k = 1, 2, 3, ··· do
    for iteration = 1, 2, 3, ··· do
        Update g^k_{φ^k} with Eq. (1).
    end for
end for
Transform {D^k} to {D̃^k} with {g^k_{φ^k}}.
for iteration = 1, 2, 3, ··· do
    for urban dynamics k = 1, 2, 3, ··· do
        Sample triplets {(X̃, X̃^{s+}, X̃^{s−})}, {(X̃, X̃^{t+}, X̃^{t−})},
        {(X̃^k, X̃^k_{p+}, X̃^{k'}_{p−})_{k'≠k}}.
    end for
    Update f_θ with Eq. (5).
end for
```

• *Temporal contrastive loss* is designed to depict the unified influence of temporal changes on multifaceted urban dynamics by identifying temporal patterns that are similar or distinct from the target observation. Considering the periodicity of urban dynamics, we utilize the Dynamic Time Wrapping (DTW) [33] to dynamically determine temporal similarity. As shown in Figure 2(a), given concatenated multifaceted urban dynamic features $\tilde{\boldsymbol{X}}$ at time t and target region R, positive samples $\tilde{\boldsymbol{X}}^{t+}$ are selected as features that exhibit a low DTW distance to $\tilde{\boldsymbol{X}}$, indicating similar temporal patterns. These positive samples can occur at nearby time steps or even at a more distant interval if they share strong temporal similarities. Conversely, negative samples $\tilde{\boldsymbol{X}}^{t-}$ are selected as those features that exhibit a high DTW distance to $\tilde{\boldsymbol{X}}$, reflecting distinct temporal dynamics. By using DTW, this temporal contrastive loss ensures that positive samples are aligned with the target's temporal structure, even if their patterns occur at shifted or irregular intervals, while negative samples represent dissimilar patterns. Therefore, the temporal

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contrastive loss is denoted as,

$$\mathcal{L}_t \triangleq \sum_{\substack{\tilde{\boldsymbol{X}}, \tilde{\boldsymbol{X}}^{t+}, \\ \tilde{\boldsymbol{X}}^{t-} \in \tilde{\mathcal{D}}}} \max(0, \|f(\tilde{\boldsymbol{X}}) - f(\tilde{\boldsymbol{X}}^{t+})\|_2^2 - \|f(\tilde{\boldsymbol{X}}) - f(\tilde{\boldsymbol{X}}^{t-})\|_2^2 + \alpha_t), \tag{3}$$

where α_t represents the margin between positive and negative temporal contrastive pairs. This approach leverages DTW's ability to align temporal sequences dynamically, making it robust to shifts or irregular periodicities in urban dynamics.

C. Multifaceted Knowledge Transfer

Given K types of urban dynamics, relying solely on the spatial and temporal contrastive losses in Eq.(2) and Eq.(3) does not enable flexible knowledge sharing or transfer across different urban dynamics. For example, it fails to capture how one type of urban dynamic influences another. To address this limitation, we design a knowledge transfer strategy by introducing a knowledge pairing contrastive loss to tackle challenge **C2**, as illustrated in Figure 2(a).

Knowledge transfer with knowledge pairing contrastive loss. The knowledge pairing contrastive loss is designed to capture and reflect the interdependencies among diverse urban dynamics data. Its primary objective is to learn how one urban dynamic influences another by leveraging the spatial and temporal context in which they co-occur. Specifically,

given a set of K urban dynamics data $\{\tilde{\mathcal{D}}^1, \cdots, \tilde{\mathcal{D}}^K\}$, we first align them based on their time and region information so that we can obtain a list of urban status that are observed at the same region R and time t, *i.e.*, $(\tilde{\boldsymbol{X}}^1, \cdots, \tilde{\boldsymbol{X}}^K)$. When learning the representation of each feature $\tilde{\boldsymbol{X}}^k$ of type k, we consider and leverage information from all other features $\{\tilde{\boldsymbol{X}}^{k'}\}$ where $k' \neq k$. We then consider positive spatial-temporal samples as those features that were observed at the same time t and region R and denote them as $\{\tilde{\boldsymbol{X}}_{p+}^{k'}\}_{k' \neq k}$, thereby highlighting their shared patterns and correlations. Negative samples are those that were observed at a different location or time where we denote as $\{\tilde{\boldsymbol{X}}_{p-}^{k'}\}_{k' \neq k}$, ensuring that the representations respect the spatial-temporal discrepancies. This approach enables the model to encode how different urban dynamics interact and influence one another. Therefore, the knowledge pairing contrastive loss for learning representations of all urban dynamic features is,

$$\mathcal{L}_p \triangleq \sum_{\substack{k=1, \\ k' \neq k}}^{K} \sum_{\substack{\tilde{\boldsymbol{X}}^k \in \tilde{\mathcal{D}}^k, \\ \tilde{\boldsymbol{X}}^{k'} \in \tilde{\mathcal{D}}^{k'}}} \max\big(0, \|f^k(\tilde{\boldsymbol{X}}^k) - f^{k'}(\tilde{\boldsymbol{X}}_{p+}^{k'})\|_2^2 - \|f^k(\tilde{\boldsymbol{X}}^k) - f^{k'}(\tilde{\boldsymbol{X}}_{p-}^{k'})\|_2^2 + \alpha_p\big). \tag{4}$$

Here, the α_p is the margin between positive and negative knowledge pairing contrastive pairs.

D. MARCEL—Multifaceted Spatial-Temporal Contrastive Learning

Structure. Given the above designs for capturing complex spatial-temporal dependencies and enabling knowledge sharing, we introduce the Multifaceted Spatial-Temporal Contrastive Learning (MARCEL) framework, as shown in Figure 2(b). The MARCEL framework begins with the pretrained preliminary representation learning module, $\{g_{\phi^k}^k\}$, which processes specific types of urban dynamics as input. The final hidden layer outputs from this module are extracted as the preliminary representations of the original urban dynamics data for each type. The subsequent representation model, denoted as f_θ, is responsible for multifaceted knowledge sharing and capturing unified spatial-temporal patterns. Positive and negative samples, represented as preliminary representations, are first passed through f_θ, enabling the refinement and alignment of spatial-temporal representations across different dynamics. The output from this model is then fed into the final feature adaptation module, which further specializes the representations, tailoring them to a specific feature type k. The final output consists of refined and task-adapted representations for each urban dynamic.

Final Objective. To train the representation model f_θ, we employ the spatial and temporal contrastive losses in Eq. (2)&(3) as well as the knowledge pairing contrastive loss introduced in Eq. (4), and try to optimize the functions by minimizing the combined loss, *i.e.*,

$$\mathcal{L}(\boldsymbol{\theta}; \tilde{\mathcal{D}}) \triangleq \gamma_1 \mathcal{L}_s(\boldsymbol{\theta}; \tilde{\mathcal{D}}) + \gamma_2 \mathcal{L}_t(\boldsymbol{\theta}; \tilde{\mathcal{D}}) + \gamma_3 \mathcal{L}_p(\boldsymbol{\theta}; \tilde{\mathcal{D}}). \tag{5}$$

Here, $\tilde{\mathcal{D}} = \{\tilde{\mathcal{D}}^k\}_{k=1}^K$, and we use γ_1, γ_2 and γ_3 as discount factors to balance the influences of the three contrastive losses. The MARCEL learning procedure in Algorithm 1.

E. Implementation for downstream tasks

After training MARCEL, we obtain latent representations that effectively capture both spatial and temporal dependencies, as well as the multifaceted intercorrelations among multiple urban dynamics. Unlike many existing studies [55], [69], [41], [8], [54], [30], [19], which directly utilize raw urban dynamics data to solve diverse urban problems—ranging from prediction to generation tasks—we instead leverage learned representations in place of raw data to perform various downstream tasks, such as traffic speed prediction, travel demand prediction, air quality prediction, and urban dynamics generation. For various urban dynamics prediction tasks (*e.g.*, traffic speed prediction, travel demand prediction, *etc*.), we use the learned representations as inputs to prediction models, which can be LSTM-based [15] or Transformer-based [46]. These models then output the final predicted urban dynamics. Beyond prediction tasks, the learned representations also contribute to urban dynamics generation tasks, such as generating detailed traffic patterns for a specific region or an entire city under given conditions, which is particularly useful for urban planning. In these tasks, we utilize our learned representations as inputs to guide the generation process. The generative models can be GAN-based [32], [59] or diffusion-based [14], [2], [26].

IV. Experiment

In our experiments, we aim to measure the effectiveness of our latent representations learned by MARCEL with the

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TABLE I: Performance (average prediction error) of three urban dynamics prediction tasks in Shenzhen, Xi'an and Chengdu.

City	Dynamics	Metrics	Models											
			MGM	PPRL	STMAE	STPT	U-BARN	RAW	STSCCL	TVRL	TCN	TimesURL	STCR	MARCEL
Shenzhen	Speed	MAPE	1.598	1.920	1.436	1.127	2.104	2.640	1.530	1.621	1.010	1.326	0.720	**0.687**
		RMSE	0.273	0.239	0.270	0.282	0.271	0.270	0.201	0.263	0.221	0.243	0.269	**0.189**
	Inflow	MAPE	1.121	0.517	1.075	0.810	0.417	0.970	1.124	0.514	0.739	1.114	1.117	**0.330**
		RMSE	0.438	0.208	0.638	0.401	0.341	0.366	0.645	0.246	0.353	0.342	0.435	**0.163**
	Demand	MAPE	0.448	0.410	0.662	0.656	0.355	0.648	0.551	0.654	0.332	0.658	0.659	**0.216**
		RMSE	0.317	0.399	0.644	0.641	0.342	0.371	0.642	0.341	0.308	0.324	0.442	**0.211**
Xi'an	Speed	MAPE	**0.165**	0.166	0.203	0.211	0.210	0.209	0.211	0.211	0.180	0.211	0.192	**0.165**
		RMSE	0.141	0.151	0.159	0.165	0.164	0.164	0.165	0.164	0.145	0.165	0.140	**0.138**
	Inflow	MAPE	0.806	0.715	0.953	1.167	1.088	0.859	1.091	1.068	0.733	1.167	0.819	**0.694**
		RMSE	0.182	0.189	0.275	0.461	0.454	0.217	0.442	0.361	0.165	0.362	0.392	**0.161**
	Demand	MAPE	0.717	0.938	0.806	0.942	1.066	0.817	0.964	0.844	0.722	0.961	0.762	**0.707**
		RMSE	0.272	0.280	0.333	0.456	0.455	0.279	0.405	0.356	0.272	0.305	0.338	**0.260**
Chengdu	Speed	MAPE	0.166	0.167	0.159	0.154	0.158	0.156	0.154	**0.153**	0.198	0.155	0.154	**0.153**
		RMSE	0.146	0.135	0.133	0.133	0.131	0.145	0.133	0.132	0.150	0.134	0.133	**0.130**
	Inflow	MAPE	0.684	0.705	0.810	0.725	0.961	0.685	1.259	1.026	0.770	1.016	0.813	**0.604**
		RMSE	0.205	0.166	0.283	0.352	0.370	0.203	0.356	0.374	0.185	0.248	0.273	**0.139**
	Demand	MAPE	1.149	1.453	1.417	1.533	1.444	1.443	1.720	1.230	1.400	1.670	1.157	**0.989**
		RMSE	0.244	0.223	0.304	0.380	0.416	0.293	0.335	0.379	0.232	0.333	0.444	**0.217**

performance of downstream tasks. We will answer the following questions with extensive experiments: **Q1:** Can our MARCEL benefit different urban dynamics prediction tasks by producing effective representations? **Q2:** Can our MARCEL benefit different urban dynamics generation tasks by producing effective representations? **Q3:** Is each component of MARCEL including the pretrained preliminary representation learning module, spatial, temporal and pairing contrastive losses effective in learning good representations? **Q4:** How do different hyperparameters affect the performance?

A. Experiment Setup

Data description. In our experiments, we collect five distinct urban dynamics datasets—travel demand, traffic speed, and traffic inflow—from multiple cities, including Chengdu, Xi'an, and Shenzhen, whereas air quality and temperature data are exclusively obtained from Shenzhen. For each city, the entire area is partitioned into equal-sized grid cells with a region size of 10×10. In Shenzhen, each urban dynamics dataset is processed to yield dimensions of $(162, 12, 63, 10, 10)$, where 162 represents the number of days, 12 corresponds to the twelve one-hour time slots per day, and 63 indicates the number of regions. For Chengdu and Xi'an, each type of urban dynamics is structured with dimensions $(30, 12, 9, 10, 10)$, with 30 days, 12 time slots per day and 9 regions.

Downstream Task Description. We implement two types of downstream tasks involving different urban dynamics: *(1) Urban dynamics prediction.* We conduct prediction tasks for various dynamics, including travel demand, traffic speed, traffic inflow, and temperature, using LSTM-based [15] models. The learned representations from MARCEL or other baselines serve as inputs. *(2) Urban dynamics generation.* We also perform urban dynamics generation tasks to generate detailed urban dynamics for regions, such as travel demand, traffic speed, and traffic inflow, under specific urban conditions. All generation tasks are implemented using cGAN-based [32] models, where the learned representations from MARCEL or state-of-the-art representation learning baselines are used as inputs.

Baselines. We evaluate MARCEL against state-of-the-art representation learning methods. *MGM* [11] adaptively guides masking in video masked autoencoders to produce effective representations. *PPRL* [41], the pretrained preliminary representation learning module in MARCEL, primarily utilizes ConvLSTM to for representation learning. *STMAE* [12] learns spatial-temporal representations by training a masked autoencoder with specialized masks. *STPT* [16] employs self-supervised learning to capture spatial-temporal dependencies in learning representations of human-generated trajectories. *U-BARN* [10] combines a spatial-temporal representation encoder with a self-supervised learning strategy to for representation learning. *STS-CCL* [23] is specifically designed for learning representations for traffic forecasting. *TVRL* [40] integrates contrastive learning and temporal embeddings in learning representations. *TCN* [39] is a self-supervised approach for learning representations and modeling temporal dependencies. *TimesURL* [25] is a self-supervised contrastive learning method designed to learn representations for time series data. *STCR* [22] captures spatial and temporal dependencies in multivariate time series with contrastive representation learning. Additionally, we compare MARCEL with a baseline that directly uses raw urban dynamics data without any representation learning, denoted as *Raw* [57].

Evaluation Metrics. We use mean absolute percentage error (MAPE) and root mean square error (RMSE) to evaluate the downstream tasks.

Experimental Settings We use the Adam optimizer with an initial learning rate of 0.0002, which is decayed by a factor of 0.1 at epochs 100, 500, and 1000. For downstream tasks, the momentum decay rates are set to 0.5 and 0.999. In Shenzhen, 50 regions are used for training and 3 of the remaining 13 regions are randomly selected for testing. For Xi'an and Chengdu, 7 regions are used for training and 2 for testing.

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TABLE II: Performance of three urban dynamics generation tasks in Shenzhen, Xi'an and Chengdu. The average errors over generated urban dynamics are provided.

City	Dynamics	Metrics	Models											
			MGM	**PPRL**	**STMAE**	**STPT**	**U-BARN**	**RAW**	**STSCCL**	**TVRL**	**TCN**	**TimesURL**	**STCR**	**MARCEL**
Shenzhen	Speed	MAPE	1.994	3.719	1.695	2.106	1.835	1.983	2.020	1.466	3.726	2.055	1.597	**1.450**
		RMSE	0.546	0.385	0.684	0.514	0.354	0.738	0.469	0.473	0.375	0.419	0.581	**0.342**
	Inflow	MAPE	1.232	1.478	0.411	1.314	0.535	0.975	1.573	0.846	1.534	1.513	0.896	**0.331**
		RMSE	0.567	0.470	0.301	0.496	0.289	0.373	0.693	0.445	0.631	0.551	0.400	**0.283**
	Demand	MAPE	0.977	0.422	0.228	0.660	0.319	0.459	0.930	0.599	1.051	0.434	0.735	**0.172**
		RMSE	0.527	0.394	0.265	0.496	0.306	0.424	0.685	0.535	0.638	0.280	0.788	**0.237**
Xi'an	Speed	MAPE	0.469	0.231	0.332	0.868	0.281	0.907	0.537	0.373	0.189	0.601	0.666	**0.177**
		RMSE	0.488	0.692	0.348	0.507	0.397	0.739	0.574	0.436	0.185	0.452	0.306	**0.179**
	Inflow	MAPE	2.190	2.504	2.612	2.205	2.410	2.236	2.217	2.222	2.824	2.455	2.195	**2.189**
		RMSE	0.729	0.860	0.689	0.789	0.611	0.778	0.735	0.822	0.694	0.647	0.720	**0.585**
	Demand	MAPE	1.741	2.014	1.658	1.706	1.772	1.819	1.632	1.675	1.656	1.806	1.601	**1.586**
		RMSE	0.529	0.818	0.776	0.831	0.601	0.841	0.805	0.811	0.501	0.595	0.594	**0.498**
Chengdu	Speed	MAPE	1.499	0.553	0.623	0.997	0.797	1.365	0.909	0.531	0.478	1.368	1.086	**0.383**
		RMSE	0.465	0.634	0.323	0.463	0.291	0.791	0.529	0.321	0.206	0.400	0.361	**0.189**
	Inflow	MAPE	1.983	2.622	2.037	2.160	2.148	2.319	2.089	2.331	2.227	2.503	2.098	**1.797**
		RMSE	0.433	0.652	0.732	0.835	0.414	0.779	0.800	0.852	0.487	0.495	0.562	**0.403**
	Demand	MAPE	2.285	2.705	2.165	2.268	2.369	2.322	2.160	2.190	2.142	2.397	2.280	**2.121**
		RMSE	0.509	0.675	0.770	0.881	0.643	0.847	0.809	0.806	0.549	0.624	0.611	**0.495**

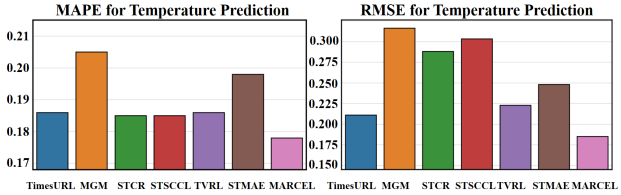


Fig. 3: Performance of urban temperature prediction in Shenzhen. The average performance is provided

B. Empirical Results

Results of Q1. To evaluate MARCEL's ability to generate effective latent representations, we first assess its impact on urban dynamics prediction tasks. As shown in Tab. I, we compare MARCEL with other representation learning baselines combined with an LSTM-based prediction model to forecast multiple urban dynamics, including traffic speed, traffic inflow, and travel demand across three cities, with the average prediction performance reported. The results show that MARCEL consistently outperforms baseline models in prediction accuracy. Notably, in Xi'an, MARCEL achieves the same MAPE for traffic speed as MGM but with a lower RMSE. Similarly, in Chengdu, MARCEL attains the same MAPE for traffic speed as TVRL but with a lower RMSE, demonstrating its stability. These findings highlight MARCEL's effectiveness in capturing complex spatial-temporal correlations and interdependencies among different urban dynamics, leading to more accurate and reliable predictions than baselines.

To further assess MARCEL's capability in generating robust representations, we extend the urban dynamics prediction task to include temperature forecasting, as illustrated in Fig. 3. In this experiment, MARCEL is trained with traffic speed, taxi inflow, travel demand, and temperature data to learn enriched representations, using latent temperature features for prediction. Results show that MARCEL consistently outperforms baselines in both MAPE and RMSE.

Results of Q2. As shown in Tab. II, we evaluate MARCEL's ability to generate effective latent representations that enhance urban dynamics generation tasks. Using a cGAN-based model, we generate multiple urban dynamics, including travel demand, traffic speed, and inflow, with their latent representations as inputs. Specifically, when generating traffic speed, the latent representations of inflow and demand serve as conditional inputs to the generative model. The results demonstrate that MARCEL consistently outperforms state-of-the-art models in both MAPE and RMSE. For instance, in Shenzhen, MARCEL achieves a similar RMSE to U-BARN in traffic speed generation while significantly reducing the MAPE, highlighting its superior ability to minimize percentage errors. Similarly, in Chengdu, MARCEL attains a comparable RMSE to U-BARN in inflow generation but substantially lowers the MAPE. Furthermore, in Xi'an, MARCEL achieves a comparable MAPE to MGM in inflow generation while significantly reducing the RMSE, demonstrating its effectiveness in minimizing absolute errors. MARCEL's advantage lies in its ability to capture complex spatial-temporal correlations and interdependencies among urban dynamics, leading to more stable and accurate generation results.

TABLE III: Impacts of different components (spatial, temporal, and pairing contrastive losses and the pretrained preliminary representations module) in MARCEL assessed on 3 urban dynamics prediction tasks in Shenzhen.

Dataset	Metric	**MARCEL**	MARCEL w/o $\mathcal{L}_p$	MARCEL w/o $\mathcal{L}_t$	MARCEL w/o $\mathcal{L}_s$	MARCEL w/o $\mathcal{L}^k_{\text{pretrain}}$
Speed	RMSE	**0.189**	0.201	0.199	0.242	0.204
	MAPE	**0.687**	0.713	0.731	0.773	0.766
Inflow	RMSE	**0.163**	0.168	0.168	0.169	0.285
	MAPE	**0.330**	0.380	0.375	0.379	0.583
Demand	RMSE	**0.211**	0.255	0.256	0.256	0.347
	MAPE	**0.216**	0.262	0.263	0.263	0.343

Results of Q3. Since our MARCEL integrates three key components in the final objective Eq. (5) including the spatial, temporal and pairing contrastive losses, in this part, we aim to evaluate the importance of each loss component by comparing MARCEL with **MARCEL w/o** $\mathcal{L}_p$ (*i.e.*, removing

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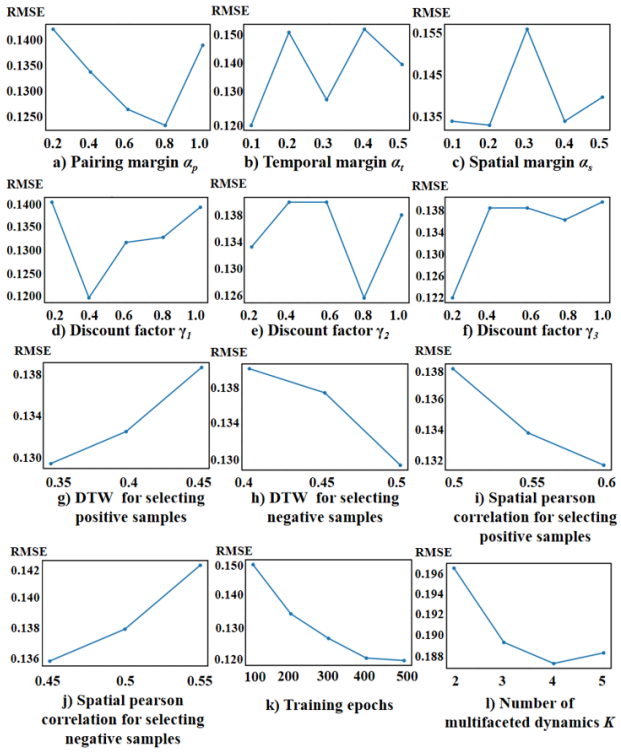


Fig. 4: Impacts of hyperparameters on the performance of traffic speed prediction.

pairing contrastive loss from Eq. (5)), **MARCEL w/o** $\mathcal{L}_s$ (*i.e.*, removing spatial contrastive loss from Eq. (5)) and **MARCEL w/o** $\mathcal{L}_t$ (*i.e.*, removing temporal contrastive loss from Eq. (5)) in urban dynamics prediction tasks. Additionally, to assess the importance of the pretrained preliminary representations, we compare MARCEL with **MARCEL w/o** $\mathcal{L}^k_{\textbf{pretrain}}$, where the original data is directly applied to Eq.(5) without pretraining. All evaluations are conducted on urban dynamics prediction tasks in Shenzhen. In this way, we can observe and measure the impact of each component on the performance and thus identify their important role in MARCEL. As shown in Tab. III, the results indicate that removing any loss component or the pretrained representations leads to an increase in both MAPE and RMSE of the final predictions. This suggests that each loss component plays a crucial role in generating high-quality representations, while the pretrained representations are essential for enhancing the model's predictive performance. With γ_1, γ_2, and γ_3 all set to 1, analysis of experimental data indicates the three components are equally important.

Results of Q4. To analyze the impact of different hyperparameters, we evaluate key hyperparameters, including the margin values α_p, α_t, and α_s; the discount factors γ_1, γ_2, and γ_3; the Dynamic Time Warping (DTW) threshold for selecting temporal positive and negative samples; spatial Pearson correlation for selecting positive and negative samples; training epochs; and the number of multifaceted dynamics K. We assess how these variations affect the learned representations and consequently the traffic speed prediction performance in Xi'an. As shown in Fig. 4(a-c), the optimal margin values for MARCEL are $\alpha_p = 0.8$, $\alpha_t = 0.1$, and $\alpha_s = 0.2$. Lower or higher values degrade performance. Similarly, Fig. 4(d-f) indicate that the best discount factors for MARCEL are $\gamma_1 = 0.4$, $\gamma_2 = 0.8$, and $\gamma_3 = 0.2$, with deviations from these values negatively impacting prediction accuracy. As shown in Fig. 4(g) and (h), a larger DTW threshold leads to a less distinct boundary for selecting positive samples, making them less representative. In contrast, increasing the DTW threshold for negative samples improves performance, as it allows for clearer separation between negative samples. Similarly, as shown in Fig. 4(i) and (j), increasing the Pearson correlation threshold for selecting spatial positive samples leads to improved performance. This is because a higher threshold ensures stronger similarity. In contrast, for negative samples, a lower correlation threshold yields better results, as it helps to distinguish negative samples more effectively. Additionally, as shown in Fig. 4(l), we evaluate how the number of multifaceted dynamics, K, affects MARCEL's learning performance. For $K = 2$, MARCEL is trained with traffic speed and taxi inflow. When $K = 3$, travel demand is added. For $K = 4$, air quality is included, and for $K = 5$, temperature is incorporated as well. The results indicate that performance generally improves as K increases, suggesting that integrating more urban dynamics provides additional spatial-temporal patterns and enhances the learning of more informative multifaceted representations.

V. Related Work

Urban dynamics prediction. Urban dynamics prediction is a multifaceted field that intersects various disciplines, including urban planning, transportation engineering, and machine learning. The primary goal is to forecast future urban status, such as traffic conditions, human mobility, and resource demands, based on historical and real-time data. Many approaches related to urban dynamics prediction are based on traditional machine learning [4], [7], [43], [29] or deep neural networks [55], [69], [41], [8], [54], [30]. For instance, cST-ML [63] and DAC-ML [57] applied meta-learning to capture the short-term traffic dynamics and guarantee good performance in traffic prediction. Some GAN-based model [62], [59], [61], [60], [64] can predict traffic status based on different urban conditions. Moreover, the incorporation of graph-based neural networks including [53], [24], [52] has further enhanced the capacity to model the complex interactions and dependencies in urban networks. These approaches leverage the inherent graph structure of urban systems, enabling more accurate and granular predictions. However, these methods are context-specific, mainly focusing on learning one aspect of urban status neglecting the correlations among different views of urban dynamics.

Contrastive learning. Contrastive learning represents a form of self-supervised learning. The core idea is to learn embeddings by bringing closer the representations of similar ('positive') pairs of data points while pushing apart the representations of dissimilar ('negative') pairs. In the field of

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computer vision, He et al. [13] introduced Momentum Contrast (MoCo) for unsupervised visual representation learning. Chen et al. [5] proposed a simple framework for contrastive learning of visual representations (SimCLR), which significantly improved the learning efficiency. In natural language processing, Logeswaran and Lee [27] proposed a contrastive approach to learn sentence embeddings. For audio processing, Oord et al. [45] introduced a contrastive predictive coding framework, showing that it is possible to learn useful representations for diverse audio tasks in an unsupervised manner. In urban domain, a lot of works [62], [38], [19], [53], [67], [59], [61], [57] try to learn representations for urban dynamics with diverse contrastive losses. However, these existing works are either not applicable in the urban domain since they fail to capture complex spatial-temporal correlations and heterogeneity, or they only consider one urban dynamics ignoring the interconnections among diverse urban dynamics.

VI. Conclusion

In conclusion, this paper tries to learn generic representations from various urban dynamics. It introduces the Multifaceted Spatial-Temporal Contrastive Learning framework (MARCEL) — a novel spatial-temporal representation model that can flexibly and comprehensively learn from various aspects of urban dynamics, and is capable of capturing complex and multifaceted spatial-temporal dependencies. MARCEL produces robust, universally applicable and adaptable representations of urban dynamics data in an innovative contrastive manner with objectives to smartly identify and capture the complicated spatial variations, temporal dynamics, and inter-dependencies across geographical regions over time. The learned representations can be easily adapted to diverse downstream applications. Extensive experiments are conducted on real-world urban dynamics datasets, the experimental results demonstrate that MARCEL outperforms other state-of-the-art baseline representation learning methods, showcasing its superior performance.

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